

PRINCE2 | P2-25

Issue Report

Formal assessment of a single issue

Meridian Group

Programme Helios — Infrastructure and Platform Stack for AI GPU Compute at Scale

Field	Detail
Document title	Issue Report
Document reference	P2-25-HEL-T2-v1.0
Project / Programme	Helios Tranche 2 — Scale Build
Version	1.0
Status	Baselined
Date	18 Sep 2026
Author	D. Okonkwo, Programme Manager
Owner	Project Manager
Approver	Change Authority or Project Board depending on impact level
Classification	Internal
Retention	7 years from programme closure

Document control

Version history

Version	Date	Author	Summary of change
0.1	04 Jun 2026	D. Okonkwo, Programme Manager	Initial draft
0.2	21 Jul 2026	D. Okonkwo, Programme Manager	Updated following workstream lead review
1.0	18 Sep 2026	D. Okonkwo, Programme Manager	Baselined at the Tranche 2 funding gate

Reviewers

Name	Role	Review scope	Date reviewed
T. Nakamura, Chief Platform Architect	Chief Platform Architect	Technical content and solution alignment	08 Sep 2026
L. Haddad, CISO	CISO	Security, assurance and residency content	10 Sep 2026
C. Whitfield, Finance Business Partner	Finance Business Partner	Cost, funding and benefit figures	11 Sep 2026
M. Verity, Head of P3O	Head of P3O	Governance standards and completeness	15 Sep 2026

Approval

This product is baselined once approval is recorded below. Any subsequent change must go through change control.

Name	Role	Decision	Date
R. Al Mazrouei, Group CTO	Senior Responsible Owner / Group CTO	Approved	18 Sep 2026
H. Farrell, Director of Infrastructure	Director of Infrastructure	Approved	18 Sep 2026
L. Haddad, CISO	CISO	Approved subject to closure of ISS-003	18 Sep 2026

Related products

Reference	Product	Relationship
P2-16	Issue Register	Register entry for this issue
P2-08	Change Control Approach	Governs the procedure
P2-19	Configuration Item Record	Affected products

How to use this template

Aspect	Guidance
Purpose	Describes, assesses and recommends a course of action for an issue that is being formally managed, providing the basis for a decision.
Framework	PRINCE2 — Created for issues requiring formal impact analysis under the Change Control Approach
Created when	When an issue is formally examined
Reviewed / updated	Updated as the issue progresses through examine, propose, decide and implement
Owner	Project Manager
Approver	Change Authority or Project Board depending on impact level
Format options	Document, presentation, register entry, or tool record — the content matters, not the medium.
Tailoring	Scale the depth of each section to the scale, risk and complexity of the work. Delete sections that add no value and record the tailoring decision in the governance approach.

This is a worked example. Every field has been completed with illustrative content for Programme Helios, a fictional AI GPU infrastructure and platform programme. All example content is shown in grey italic — the black text is the template itself. Replace the grey italic content with your own.

Quality criteria

Use these to check the product is fit for purpose before submitting it for approval.

- The issue is described so that someone unfamiliar with it can understand it
- Impact analysis covers all six performance dimensions and affected products
- The recommendation follows logically from the analysis
- The decision and its rationale are recorded, including for rejected issues

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1. Issue identification

Field	Detail
Issue reference	ALM tool — Helios ART board
Issue title	Helios Tranche 2 — Scale Build
Issue type	[Request for change / Off-specification / Problem or concern]
Date raised	18 Sep 2026
Raised by	P. Raghavan, Platform Engineering Lead
Stage	Tranche 2 — Scale build
Priority	Should have
Severity	Delegated to the Programme Manager within the agreed tolerance
Current status	Baselined

2. Description

What the issue is, when and how it was identified, and what has happened so far.

In scope: the physical infrastructure in DXB-1 halls A and B, the compute and storage estate, the network fabric, the orchestration and scheduling platform, the MLOps and developer experience layer, the security controls required for multi-tenancy, and the operational readiness required to run the estate. Out of scope: the models themselves, business-unit data engineering, the AUH-2 second site (Tranche 3), and any change to the existing corporate WAN.

3. Impact analysis

Impact on project performance

Dimension	Impact if not addressed	Impact of proposed action
Time	4 weeks	90 days
Cost	AED 118,000,000 for Tranche 2; tolerance of +5% before escalation	AED 118,000,000 for Tranche 2; tolerance of +5% before escalation
Scope	Defined by the Tranche 2 scope statement; changed only by approved change request	<i>In scope: the physical infrastructure in DXB-1 halls A and B, the compute and storage estate, the network fabric, the orchestration and scheduling platform, the MLOps and developer experience layer, the security controls required for multi-tenancy, and the operational readiness required to run the estate. Out of scope: the models themselves, business-unit data engineering, the AUH-2 second site (Tranche 3), and any change to the existing corporate WAN.</i>
Quality	No open critical or high security findings; acceptance benchmarks met in full	No open critical or high security findings; acceptance benchmarks met in full

Dimension	Impact if not addressed	Impact of proposed action
Risk	No individual risk may remain above a residual score of 12 without SRO acceptance	No individual risk may remain above a residual score of 12 without SRO acceptance
Benefit	No more than 10% variance against the benefit profile before escalation	No more than 10% variance against the benefit profile before escalation

Products affected

Product ID	Product	Nature of impact	Re-work required
Hall A accepted at 1,024 GPUs with acceptance certificate	Independent security assurance report with findings closed	Scheduled fraud workloads could be pre-empted by research jobs	Yes
Independent security assurance report with findings closed	Operations runbook set with two completed live drills	Design change after the structural order would require rework	No
Operations runbook set with two completed live drills	Parallel filesystem accepted at 1.2 TB/s sustained	Chargeback introduces cost where compute was previously free	Partially
Parallel filesystem accepted at 1.2 TB/s sustained	400G fabric commissioned and NCCL-validated	Departmental autonomy is removed with no compensating guarantee	Yes

Wider impact

Effect on other projects, operational services, suppliers, contractual positions, compliance obligations or stakeholder commitments.

4. Options considered

Option	Description	Cost	Time	Risk	Pros / cons
Do nothing	Multi-tenant orchestration with fair-share quota across AI Research, Fraud Analytics and Retail	AED 17,800,000	3 working days	Weak tenant isolation exposes model weights across business units (RSK-	Applies from hall A handover; hall B follows in Tranche 2 closure

Option	Description	Cost	Time	Risk	Pros / cons
				003)	
Option B — accept 48 kW density and cap at 1,536 GPUs	400G RoCEv2 leaf-spine fabric tuned for collective communication at 256-GPU job scale	AED 1,120,000	2 hours	Scarce platform engineering skills extend contractor dependence (RSK-004)	Maintained by the P3O as part of the assurance evidence library
Option A — add a second CDU to hall A	Liquid-cooled racks at 60 kW with rear-door heat exchangers and CDU redundancy at N+1	AED 91,200,000	60 minutes	Fabric tuning falls short of the 92% all-reduce target (RSK-005)	Enforced automatically in the delivery pipeline, with evidence retained
Do nothing	Consolidate all AI training workloads onto a shared 2,048-GPU platform in DXB-1 halls A and B	AED 4,200,000	90 days	Slow workload migration leaves the platform under-used (RSK-015)	Dependent on the Enterprise Identity Programme (DEP-003)

5. Recommendation

The recommended option, the rationale, and what is required to implement it.

Proceed with Tranche 2 at the full 2,048-GPU configuration, subject to two conditions: written confirmation of vendor allocation before the structural order is placed, and closure of the outstanding critical penetration test finding (ISS-003) before multi-tenant go-live.

6. Decision

Field	Detail
Decision	[Approve / Reject / Defer / Approve with concession]
Decision authority	D. Okonkwo, Programme Manager
Decision maker	Approved
Date of decision	18 Sep 2026
Rationale	Physical infrastructure has long lead times and low requirement volatility; the platform layer has short feedback loops and genuinely emergent requirements
Change budget drawn	AED 96,000
Conditions	Performance bond of 10%, step-in rights, and design documentation held in escrow

7. Implementation

Action	Owner	Target date	Status	Completed
Secure written migration commitments from three business units	D. Okonkwo, Programme Manager	31 Jan 2027	Complete	Yes
Retune CDU set-points and model a secondary heat rejection loop	A. Kowalski, Data Centre Lead	15 Dec 2026	Complete	No
Broker all vendor access through the monitored in-country jump host	S. Bhatt, Project Manager	30 Nov 2026	Complete	Yes
Confirm vendor allocation in writing before placing the structural order	L. Haddad, CISO	15 Oct 2026	Complete	Partially
Enforce mutual TLS on the cluster API and retest by 22 Sep 2026	A. Kowalski, Data Centre Lead	18 Sep 2026	In progress	Yes

8. Closure

Field	Detail
Closure verified by	G. Lindqvist, Service Operations Lead
Date closed	18 Sep 2026
Baselines updated	18 Sep 2026
Lesson recorded?	Configuration management database